

1st Semester
2026-2027

CS 314 – METHODS OF RESEARCH



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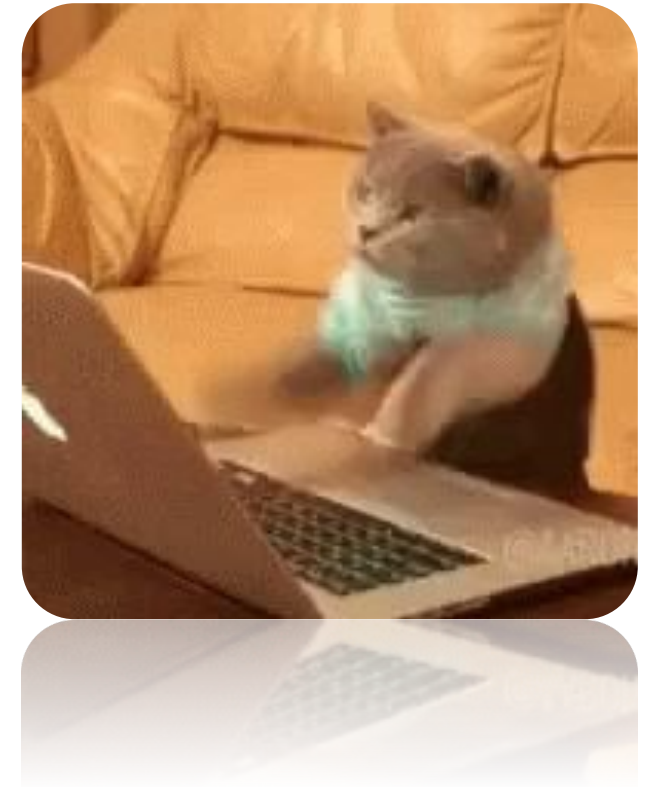
INTRODUCTION TO RESEARCH IN COMPUTER SCIENCE

Objectives

- Define research and explain its role in advancing computer science
- Classify different types of computer science research through examples
- Defend the value of research in solving real-world computing problems

WHAT IS RESEARCH

- The **systematic investigation** into and study of **materials** and sources in order to **establish facts** and **reach new conclusions** (Oxford)
- **Investigation** or **experimentation** aimed at the **discovery** and interpretation of facts, revision of accepted theories or laws in the light of new facts, or practical application of such new or revised theories or laws (Merriam-Webster)
- Then what is research in the context of computer science?



RESEARCH IN COMPUTER SCIENCE

- A **systematic and rigorous process** of inquiry aimed at generating new knowledge or advancing technology within computing
- It entails the formal formulation of a **problem, designing and conducting experiments or constructs, analyzing outcomes, and evaluating findings** with the goal of contributing to **theory, algorithms, systems, or applications**



TYPES OF RESEARCH IN COMPUTER SCIENCE

- Theoretical research
- Applied research
- Experimental research
- Human-centered research
- Emerging and Interdisciplinary research

THEORETICAL RESEARCH IN COMPUTER SCIENCE

- Focuses on foundations of computing
- Key topics include:
 - Algorithm design and analysis
 - Data Structures
 - Automata theory
 - Information theory
 - Computational complexity theory
 - Cryptography
 - Formal methods
 - Artificial Intelligence/Machine Learning

THEORETICAL RESEARCH IN COMPUTER SCIENCE

Breaking the Sorting Barrier for Directed Single-Source Shortest Paths

Ran Duan^{*} Jiayi Mao^{*} Xiao Mao[†] Xinkai Shu[‡] Longhui Yin^{*}

July 31, 2025

Abstract

We give a deterministic $O(m \log^{2/3} n)$ -time algorithm for single-source shortest paths (SSSP) on directed graphs with real non-negative edge weights in the comparison-addition model. This is the first result to break the $O(m + n \log n)$ time bound of Dijkstra's algorithm on sparse graphs, showing that Dijkstra's algorithm is not optimal for SSSP.

1 Introduction

In an m -edge n -vertex directed graph $G = (V, E)$ with a non-negative weight function $w : E \rightarrow \mathbb{R}_{\geq 0}$, single-source shortest path (SSSP) considers the lengths of the shortest paths from a source vertex s to all $v \in V$. Designing faster algorithms for SSSP is one of the most fundamental problems in graph theory, with exciting improvements since the 50s.

The textbook Dijkstra's algorithm [Dij59], combined with advanced data structures such as the Fibonacci heap [FT87] or the relaxed heap [DGST88], solves SSSP in $O(m + n \log n)$ time. It works in the *comparison-addition* model, natural for real-number inputs, where only comparison and addition operations on edge weights are allowed, and each operation consumes unit time. For undirected graphs, Pettie and Ramachandran [PR05] proposed a hierarchy-based algorithm which runs in $O(m\alpha(m, n) + \min\{n \log n, n \log \log r\})$ time in the comparison-addition model, where α is the inverse-Ackermann function and r bounds the ratio between any two edge weights.

Dijkstra's algorithm also produces an ordering of vertices by distances from the source as a byproduct. A recent contribution by Haeupler, Hladík, Rozhoň, Tarjan and Tětek [HHR⁺24] showed that Dijkstra's algorithm is optimal if we require the algorithm to output the order of vertices by distances. If only the distances and not the ordering are required, a recent result by Duan, Mao, Shu and Yin [DMSY23] provided an $O(m\sqrt{\log n \log \log n})$ -time randomized SSSP algorithm for undirected graphs, better than $O(n \log n)$ in sparse graphs. However it remains to break such a sorting barrier in directed graphs.

APPLIED RESEARCH IN COMPUTER SCIENCE

- Aims to solve practical problems using computer science principles
- Key topics include:
 - Software Engineering
 - Computer networks and distributed systems
 - Cloud computing
 - Embedded systems
 - Operating systems development
 - Computer Vision
 - Natural Language Processing

APPLIED RESEARCH IN COMPUTER SCIENCE

Design and Development of Web-based and Mobile Water Billing System

Alvin T. Remolado, Alme M. Aparicio, Daryl B. Valdez, Rey Anthony G. Godmalin, Ariel Christian C. Viodor, Ramil H. Lim, Ruvel I. Abad, Jai Gelle J. Aaron, Janeth E. Banol, Angela Evian Mae T. Claro

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ABSTRACT

The water billing system of the Local Government Unit of Clarin is currently done through a manual process, which often causes delays, errors, and inconvenience to both staff and consumers. This study aimed to design and develop a web-based and mobile water billing system that improves efficiency, accuracy, and transparency in billing and payment collection. The system was developed using Next.js for the web platform, Android Studio with Kotlin and Jetpack Compose for the mobile application, and Google Firebase for real-time data storage and synchronization. The functional prototype allowed meter readers to record water consumption directly through the mobile application, while the web application provided staff and administrators with tools for billing, payment monitoring, and customer management. Results showed that the proposed system is faster, more accurate, and more reliable compared to the manual system. It reduced human error, improved the timeliness of bill issuance, and offered flexible payment options. The study concludes that the developed system can modernize the water billing process of the LGU of Clarin and serve as a model for other local government units that face similar challenges.

APPLIED RESEARCH IN COMPUTER SCIENCE

MangroveLens: A Smart Solution for Mangrove Species Identification Through MobileNetV3 Network Architecture and Biodiversity Monitoring

Publisher: IEEE

[Cite This](#)

[PDF](#)

Ariel Christian C. Viodor [All Authors](#)

1

Cites in
Paper

83

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Abstract:

Mangroves, as vital coastal ecosystems, play a significant role in maintaining biodiversity and providing essential ecosystem services. However, difficulties with species identification and biodiversity monitoring impede their conservation. In this work, we provide MangroveLens, a smart solution for biodiversity monitoring and mangrove species identification that makes use of MobileNetV3. We suggest utilizing mobile device images to recognize mangrove species using a deep learning-based method. MobileNetV3, renowned for its accuracy and efficiency, is used because it works well in contexts with limited resources. We exhibit the effectiveness of MangroveLens by conducting extensive experiments on real-world mangrove datasets, demonstrating its capacity to precisely identify different species of mangroves and monitor their biodiversity. Streamline biodiversity monitoring with an easy-to-use interface, real-time data documentation, and a cloud-based setup for viable analysis. Our findings suggest that MangroveLens provides a useful and effective tool for policymakers, academics, and conservationists to monitor and manage mangrove ecosystems, enabling focused conservation efforts and improving our knowledge of these important habitats.

Published in: 2024 IEEE Open Conference of Electrical, Electronic and Information Sciences (eStream)

EXPERIMENTAL RESEARCH IN COMPUTER SCIENCE

- Involves building and testing prototypes, simulations, or systems to validate hypotheses
- Analyzes real-world data or user behavior to derive insights
- Key topics include:
 - Machine Learning model development and evaluation
 - Robotics and sensor-based systems
 - Computer Vision
 - Natural Language Processing

EXPERIMENTAL RESEARCH IN COMPUTER SCIENCE

RESEARCH-ARTICLE X in   

Habitat Prediction and Knowledge Extraction for Marine Bivalves using Machine Learning Techniques

Authors:  [Alme Bautista Maravillas](#),  [Larmie Santos Feliscuzo](#),  [James Arnold Estrellosa Nogra](#) | [Authors Info & Claims](#)

[ISMSI '23: Proceedings of the 2023 7th International Conference on Intelligent Systems, Metaheuristics & Swarm Intelligence](#)
Pages 28 - 35 • <https://doi.org/10.1145/3596947.3596964>

Published: 27 July 2023 [Publication History](#) 

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Compare the performance of four machine learning models to predict species distribution of marine bivalves

EXPERIMENTAL RESEARCH IN COMPUTER SCIENCE

Mangrove Species Identification Using Deep Neural Network

Publisher: IEEE

[Cite This](#)

[PDF](#)

Ariel Christian C. Viodor ; Chris Jordan G. Aliac ; Larmie T. Santos-Feliscuzo [All Authors](#)

1
Cites in
Paper

165
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Abstract:

Mangroves play an essential component in the coastal ecosystem and the community. Identifying mangrove plant species is critical for biodiversity assessment, restoration, or conservation programs apart from ecosystem protection from anthropogenic activities. Yet, comprehensive data for managing coastal ecosystems sustainably are being gathered manually. Furthermore, identifying mangrove species by traditional keys is complicated, time-consuming, and frustrating due to technical terminology for non-experts. The growing need to identify mangroves in a timely and cost-effective manner motivates our study to create an automated approach for classifying the species from images of mangrove leaves. Recent advances in deep learning algorithms have resulted in studies demonstrating excellent performance identification tasks for plant species. This paper examines existing deep neural network-based machine learning applications for identifying plants designed for hand-held device usage. Driven by low-cost computation, size, and time requirements, we propose the use of MobileNet architecture and transfer learning for leaf-based plant recognition. We provide a dataset of images comprising five (5) several types of mangrove species. Then, we perform experimental validation on the proposed state-of-the-art architecture and transfer learning. Our results show that accuracy up to 97.07% was achieved using MobileNetV3Small and transfer learning for mangrove species identification proving the viability of its use in mobile applications. The saved model has a low-cost computation that runs within a smartphone application.

Published in: 2022 6th International Conference on Information Technology, Information Systems and Electrical Engineering (ICITISEE)

EXPERIMENTAL RESEARCH IN COMPUTER SCIENCE

Cacao Pod Infection Level Classification Using Transfer Learning

Publisher: IEEE

[Cite This](#)

[PDF](#)

[Rey Anthony Godmalin](#) ; [Chris Jordan Aliac](#) ; [Larmie Feliscuzo](#) [All Authors](#)

1
Cites in
Paper

172
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Abstract:

This research aims to address the constant threat to cacao farming, particularly from diseases like black pod rot, by utilizing Artificial Intelligence (AI) and Deep Learning algorithms. An experimental research design method was used to train a convolutional neural network that can classify three levels of cacao pod infection: low, moderate, and severe. Under controlled conditions, the model achieved an accuracy of 91% in correctly classifying the cacao pod condition. The study recommends further research to integrate the model with hardware monitoring and surveillance devices to enable real-time classification of cacao pod conditions in the actual field. With this capability, farmers can respond quickly and effectively to mitigate production loss caused by cacao diseases. This research presents a promising approach to leveraging AI in cacao farming, with potential benefits for farmers, researchers, and the broader community.

Published in: 2023 IEEE Open Conference of Electrical, Electronic and Information Sciences (eStream)

EXPERIMENTAL RESEARCH IN COMPUTER SCIENCE

Comparative Analysis of Random Forest and Support Vector Classifier for Student Risk Assessment: A Case Study of Colegio de Getafe

Publisher: IEEE

[Cite This](#)

[PDF](#)

Alvin T. Remolado ; Deborah G. Brosas [All Authors](#)

29

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Abstract:

This research aims to compare the random forest classifier and support vector classifier to the survey data of the risk assessment system for Colegio de Getafe (CDG), a higher educational institution, to identify at-risk students and provide necessary support and interventions. The current manual data collection and analysis is time-consuming and often leads to inaccurate assessments. This study uses machine learning algorithms to analyze student survey data and classify students based on their unique characteristics, such as the differently abled, at-risk, and special needs. This study involves data collection, preprocessing, cleaning, transformation, and modeling using Random Forest and Support Vector Classifier algorithms. Evaluation metrics include accuracy, precision, recall, and F1 score. The results indicate that the Support Vector Classifier has an accuracy of 100%, while the Random Forest Classifier shows an accuracy of 98.09%. The SVC's ability to capture complex relationships and decision boundaries in the data contributes to its superior performance. The findings highlight the potential of the SVC as a powerful tool for analyzing survey data in higher education and can inform decision-making and targeted interventions by school administrators. Further research can explore the implications of different data mining techniques and parameter configurations on similar datasets.

Published in: 2023 Sixth International Conference on Vocational Education and Electrical Engineering (ICVEE)

EXPERIMENTAL RESEARCH IN COMPUTER SCIENCE

Medicinal Plant Classification using Convolutional Neural Network and Transfer Learning

Publisher: IEEE

[Cite This](#)

[PDF](#)

Daryl B. Valdez ; Chris Jordan G. Aliac ; Larmie S. Feliscuzo [All Authors](#)

14
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Abstract:

Medicinal plants are not only an essential source of therapeutic compounds but also an alternative source of medications used by most people around the world. Due to recent advances in computer vision, plant identification from images has become a rapidly developing research field. Various results showed good accuracy, precision, and real-world applications. This paper aimed to investigate an accurate and precise automated identification of medicinal plants. We present a new medicinal plant dataset containing images of ten (10) classes of medicinal plant species and one (1) class containing a mixed variety of weeds, vines, and non-medicinal plants. Then we proposed a model based on MobileNetV3 architecture for a low-cost, reliable, and efficient medicinal plant classification. Using the proposed model and Transfer Learning, results revealed a 97.43% accuracy on the challenging task. Overall, the findings revealed the feasibility of an efficient and reliable medicinal plant classifier for real-world applications.

Published in: 2022 IEEE International Conference on Artificial Intelligence in Engineering and Technology (IICAET)

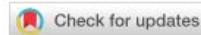
EXPERIMENTAL RESEARCH IN COMPUTER SCIENCE

A Deep Learning Approach of Recognizing Natural Disasters on Images using Convolutional Neural Network and Transfer Learning

Authors:  Daryl B. Valdez,  Rey Anthony G. Godmalin [Authors Info & Claims](#)

ICARTI '21: Proceedings of the International Conference on Artificial Intelligence and its Applications
Article No.: 4, Pages 1 - 7 • <https://doi.org/10.1145/3487923.3487927>

Published: 09 December 2021 [Publication History](#)



 16  240



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Abstract

Natural disasters are uncontrollable phenomena occurring yearly which cause extensive damage to lives, property and cause permanent damage to the environment. However by, using Deep Learning, real-time recognition of these disasters can help the victims and emergency response agencies during the onset of these destructive events. At present, there are still gaps in the literature regarding real-time natural disaster recognition. In this paper, we present a dataset for the joint classification of natural disasters and intensity. We also proposed a lightweight convolutional neural network with two classification heads for the two tasks. This study leveraged on transfer learning in training the network to recognize natural disasters, as well as detecting normal, no-disaster images. At the same time, it is also capable of recognizing disaster intensity. Under controlled conditions, the model showed promising results on the two classification tasks. Thus, the study proved that accurate recognition of natural disasters is possible using a lightweight model and transfer learning. We hope that this study would lead to development of monitoring or surveillance systems that can perform accurate, on-the-ground, and real-time recognition of natural disasters allowing for rapid emergency responses mitigating the loss of lives and damages to properties.

EXPERIMENTAL RESEARCH IN COMPUTER SCIENCE

Optimizing MobileNetV3 Using Particle Swarm Optimization Algorithm for the Recognition of Bean Leaf Diseases

Publisher: **IEEE**

[Cite This](#)

[PDF](#)

Daryl B. Valdez [All Authors](#)

1
Cites in
Paper

48
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Abstract:

Crops play a vital role in human nutrition and overall well-being. Beans are an economically significant type of crops that not only provide a rich source of protein but also offers substantial health benefits. However, beans are susceptible to bacterial and fungal infections. If left uncontrolled, these diseases could result in severe consequences. Diagnosis of the affected leaves can be done by plant pathologists, which is not readily available in local and remote areas. Using deep learning algorithms, an accurate classification of bean leaf diseases can offset this problem from an early stage. This research introduces a framework for optimizing MobileNetV3 using Particle Swarm Optimization that can accurately identify healthy or diseased bean leaves from images captured in real-world conditions. With having only 2.9M parameters, the best model achieved a high degree of accuracy at 98.44% on unseen data, outperforming the models proposed by other researchers on the same classification task. Results suggest that the model can be adopted in the field to increase the efficiency in the early identification of bean leaf diseases. Lastly, the same methodology can also be applied in solving plant leaf diseases from other economically significant crops.

Published in: [2024 IEEE Open Conference of Electrical, Electronic and Information Sciences \(eStream\)](#)

EXPERIMENTAL RESEARCH IN COMPUTER SCIENCE

Optimized Fruit Detection in Complex Environments Using YOLOv11n for Smart Agricultural Applications

Publisher: IEEE

[Cite This](#)

[PDF](#)

Daryl B. Valdez [All Authors](#)

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Abstract:

This study explores the application of the YOLOv11 object detection algorithm for precise and efficient fruit detection in complex image environments. The research focuses on leveraging YOLOv11's advanced single-stage detection architecture and Transfer Learning to enable real-time fruit identification while maintaining high detection accuracy, particularly for edge computing scenarios where computational efficiency is essential. To evaluate the effectiveness of the proposed system, the model was trained on a specialized fruit dataset consisting of images taken from real-world agricultural settings. The dataset comprises eleven distinct fruit classes captured under diverse conditions, including varying lighting, occlusions, and different angles. Experimental results revealed that the model achieved a mean average precision (mAP) of 90%, demonstrating its ability to localize and classify fruits with a high degree of reliability. The findings of this study have significant implications for the agricultural sector, particularly in automation-driven applications such as fruit harvesting, post-harvest processing, quality assessment, and inventory management. By integrating YOLOv11-based detection systems in the process, farms and agribusinesses can improve efficiency, reduce labor costs, and enhance productivity.

Published in: [2025 IEEE Open Conference of Electrical, Electronic and Information Sciences \(eStream\)](#)

HUMAN-CENTERED RESEARCH IN COMPUTER SCIENCE

- Involves building and testing prototypes, simulations, or systems to validate hypotheses
- Analyzes real-world data or user behavior to derive insights
- Key topics include:
 - Usability studies (software usability testing)
 - Human Computer Interaction
 - User Experience research (interaction design, accessibility, virtual/augmented reality)

Design and evaluation of AR-based adaptive human-computer interaction cognitive training

Man Chu ^{a b}, Jing Qu ^{a b}, Tan Zou ^{a b}, Qinbiao Li ^c, Lingguo Bu ^{a b}  , Yiran Shen ^b

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<https://doi.org/10.1016/j.ijhcs.2025.103504> 

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Abstract

As human-computer interaction (HCI) technology advances, the use of augmented reality (AR) in cognitive training is becoming more prevalent. However, traditional training methods often apply a one-size-fits-all approach, failing to accommodate the varied training needs of individuals with different cognitive levels. Additionally, most HCI systems use subjective questionnaires for evaluation, which can be influenced by the subjects' emotional and mental states. To overcome these challenges, this study developed an AR-based adaptive HCI cognitive training system that dynamically adjusts task difficulty based on real-time user performance. We used multi-source data to empirically validate the effectiveness of adaptive HCI in cognitive training. Specifically, we recorded functional Near-Infrared Spectroscopy (fNIRS) data, movement data, task performance, and subjective feedback from 22 elderly participants, dividing them into two groups—low cognitive group and normal cognitive group. The results showed that the system exerted a significant influence on brain functional connectivity (FC) associated with cognition, movement, and vision. Changes in FC may highlight the benefits of adaptive HCI training strategies. Furthermore, participants with normal cognitive abilities significantly outperformed their low cognitive counterparts in task performance. In conclusion, this study designed and evaluated an AR-based adaptive HCI cognitive training system that ensures personalized training. It demonstrated the feasibility of adaptive HCI strategies in cognitive rehabilitation by incorporating physiological and behavioral data, thereby enhancing the precision of quantitative assessments for HCI systems.

DigCode—A generic mid-air gesture coding method on human-computer interaction

Xiaozhou Zhou ^a  , Lesong Jia ^{a b}, Ruidong Bai ^a, Chengqi Xue ^a

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<https://doi.org/10.1016/j.ijhcs.2024.103302> 

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Abstract

With high flexibility and rich semantic expressiveness, mid-air gesture interaction is an important part of natural human-computer interaction (HCI) and has broad application prospects. However, there is no unified representation frame for designing, recording, investigating and comparing HCI mid-air gestures. Therefore, this paper proposes an interpretable coding method, DigCode, for HCI mid-air gestures. DigCode converts the unstructured continuous actions into structured discrete string encoding. From the perspective of human cognition and expression, the research employed psychophysical methods to divide gesture actions into discrete intervals, defined the coding rules of representation in letters and numbers, and developed automated programs to enable encoding and decoding by using gesture sensors. The coding method can cover the existing representations of HCI mid-air gestures by considering human understanding and computer recognition and can be applied to HCI mid-air gesture design and gesture library construction.

EMERGING AND INTERDISCIPLINARY RESEARCH IN COMPUTER SCIENCE

- Combines computer science with other fields to address domain-specific challenges.
- Investigates emerging or uncharted areas to identify new possibilities.
- Key topics include:
 - Emerging Researches
 - Quantum Computing
 - Artificial General Intelligence, LLMs, Agentic AI systems
 - Blockchain and decentralized systems
 - Explainable AI
 - AI Ethics
 - Interdisciplinary Researches
 - Bioinformatics
 - Environmental modelling
 - AI in education (educational technology)

Design of an efficient fault-tolerant quantum-computing circuit with quantum neural network learning

Chen Lin ^a ✉, Rucong Xu ^a ✉, Yun Li ^{a b}  ✉

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<https://doi.org/10.1016/j.engappai.2025.110808> ↗

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Fault-tolerance is key to the practical realization of quantum computation, but the design of an efficient, low-overhead and fault-tolerant error-correction circuit remains a major challenge so far. To help address this issue, we first propose a noise-adaptive dissipative quantum neural network (DQNN) model to mitigate the effects of error propagation for constructing a fault-tolerant quantum circuit. Then, we develop a method for preparing a fault-tolerant auxiliary entangled state based on the DQNN model, reducing computational delays and qubit resource consumption. This method utilizes the adaptability of quantum machine learning to the distribution of noisy inputs and its practicality in noisy intermediate scale quantum devices, thus reducing interaction with classical computers and further optimizing the real-time requirements for active error-correction. By integrating quantum error-correction and quantum neural network learning, this DQNN scheme provides a novel solution for constructing scalable fault-tolerant quantum computation. Compared with existing fault-tolerant methods, the DQNN process requires fewer error-propagation efforts and offers higher fidelity in a noisy environment for error thresholds higher than 10^{-4} . The effectiveness of this method is verified through experimental simulations using the Qiskit. The code for experiments and model in this paper can be found on GitHub:

https://github.com/Ricardo-Vv/Qiskit_exam/tree/master ↗.

EMERGING RESEARCH IN COMPUTER SCIENCE

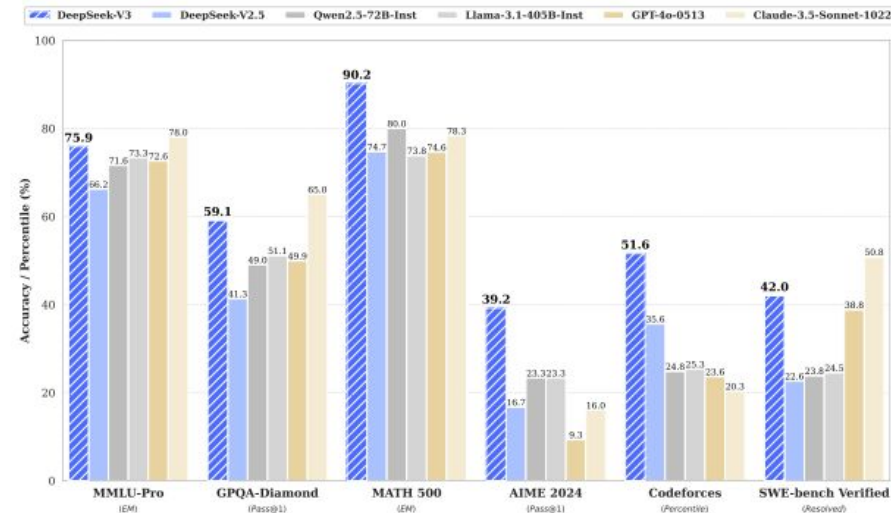
DeepSeek-V3 Technical Report

DeepSeek-AI

research@deepseek.com

Abstract

We present DeepSeek-V3, a strong Mixture-of-Experts (MoE) language model with 671B total parameters with 37B activated for each token. To achieve efficient inference and cost-effective training, DeepSeek-V3 adopts Multi-head Latent Attention (MLA) and DeepSeekMoE architectures, which were thoroughly validated in DeepSeek-V2. Furthermore, DeepSeek-V3 pioneers an auxiliary-loss-free strategy for load balancing and sets a multi-token prediction training objective for stronger performance. We pre-train DeepSeek-V3 on 14.8 trillion diverse and high-quality tokens, followed by Supervised Fine-Tuning and Reinforcement Learning stages to fully harness its capabilities. Comprehensive evaluations reveal that DeepSeek-V3 outperforms other open-source models and achieves performance comparable to leading closed-source models. Despite its excellent performance, DeepSeek-V3 requires only 2.788M H800 GPU hours for its full training. In addition, its training process is remarkably stable. Throughout the entire training process, we did not experience any irrecoverable loss spikes or perform any rollbacks. The model checkpoints are available at <https://github.com/deepseek-ai/DeepSeek-V3>.



GPT-4 Technical Report

OpenAI*

Abstract

We report the development of GPT-4, a large-scale, multimodal model which can accept image and text inputs and produce text outputs. While less capable than humans in many real-world scenarios, GPT-4 exhibits human-level performance on various professional and academic benchmarks, including passing a simulated bar exam with a score around the top 10% of test takers. GPT-4 is a Transformer-based model pre-trained to predict the next token in a document. The post-training alignment process results in improved performance on measures of factuality and adherence to desired behavior. A core component of this project was developing infrastructure and optimization methods that behave predictably across a wide range of scales. This allowed us to accurately predict some aspects of GPT-4's performance based on models trained with no more than 1/1,000th the compute of GPT-4.

INTERDISCIPLINARY RESEARCH IN COMPUTER SCIENCE

Predicting Geographic Distribution and Potential Habitat of Marine Bivalves *

Publisher: IEEE

Cite This

PDF

Alme B. Maravillas  [All Authors](#)

52

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Abstract:

Understanding the habitat suitability of the species has been one of the main focus of biodiversity conservation. Species Distribution Modelling (SDM) has great potential to support marine conservation planning. SDM can forecast the optimal conditions for species cultivation, aiding in the prevention of habitat loss and biodiversity degradation. Four habitat suitability models-DNN, MaxEnt, RF, and GBM-were utilized to forecast the distribution of marine bivalves. Also, a stacking ensemble-based estimation model was developed, utilizing the model performance outcomes as input to enhance estimation accuracy. Finally, the habitat suitability results based on the performance evaluation metrics such as the area under the curve (AUC), sensitivity, specificity, the kappa statistic, and the TSS were used to evaluate the predictive performance of the SDMs. The experimentation results indicate that the Ensemble model delivers superior predictive performance as evidenced by its exceptional scores and demonstrated strong predictive capabilities in forecasting the potential habitat of bivalve species. The implemented model achieves impressive AUC values of 0.98, along with Kappa statistics and Specificity scores of 0.96, and Sensitivity and TSS scores of 0.97. These findings underscore the effectiveness of the Ensemble model in accurately predicting bivalve habitat suitability, thus providing valuable insights for marine conservation planning efforts.

Published in: 2024 IEEE Open Conference of Electrical, Electronic and Information Sciences (eStream)

PROCESS OF CONDUCTING RESEARCH IN COMPUTER SCIENCE

